Tswalu Steel Fabrication

Network Design Document

Dual-Stack IPv4/IPv6 VLAN Network – Cisco Packet Tracer

Purpose

This document describes the final network design implemented in Cisco Packet Tracer. The design uses one Cisco router, one switch, departmental VLANs, router-on-a-stick inter-VLAN routing, IPv4 and IPv6 addressing, and a separate wireless guest network.

Final Physical/Logical Components

Component	Role
Router R1	Inter-VLAN routing and IPv4/IPv6 default gateways
Switch SW1	VLAN segmentation and 802.1Q trunking
PC-Admin	Management VLAN endpoint
PC-Production	Production VLAN endpoint
PC-Finance	Finance VLAN endpoint
PC-IT	IT services VLAN endpoint
PC-Office	IT services VLAN endpoint
SRV1	Server on IT services VLAN
AP-Guest	Wireless access point for guest network
Laptop-Guest	Wireless guest endpoint

VLAN and Addressing Design

VLAN	Name	IPv4 Network	IPv4 Gateway	IPv6 Network	IPv6 Gateway
10	MANAGEMENT	172.30.12.64/27	172.30.12.65	2001:DB8:ACAD:10::/64	2001:DB8:ACAD:10::1
20	PRODUCTION	172.30.12.0/26	172.30.12.1	2001:DB8:ACAD:20::/64	2001:DB8:ACAD:20::1
30	FINANCE	172.30.12.128/28	172.30.12.129	2001:DB8:ACAD:30::/64	2001:DB8:ACAD:30::1
40	IT_SERVICES	172.30.12.144/28	172.30.12.145	2001:DB8:ACAD:40::/64	2001:DB8:ACAD:40::1
50	GUEST	172.30.12.96/27	172.30.12.97	2001:DB8:ACAD:50::/64	2001:DB8:ACAD:50::1

Network Architecture

The switch connects end devices using access ports. GigabitEthernet0/1 on SW1 is configured as an 802.1Q trunk and carries VLANs 10, 20, 30, 40 and 50 to GigabitEthernet0/0 on R1. R1 uses subinterfaces .10, .20, .30, .40 and .50 as the default gateways for the VLANs.

Router-on-a-Stick Design

Router R1 provides inter-VLAN routing through 802.1Q subinterfaces. IPv6 unicast routing is enabled so that IPv6 traffic can be routed between VLANs.

Switching and Trunking

SW1 has access ports assigned to the required VLANs. The final observed port assignments were Fa0/1→VLAN 10, Fa0/2→VLAN 20, Fa0/3→VLAN 30, Fa0/4–Fa0/5→VLAN 40, and Fa0/6→VLAN 50. Gi0/1 is the trunk to R1 and allows VLANs 10,20,30,40,50.

Wireless Guest Network

The guest wireless laptop uses the WPC300N wireless adapter and connects through AP-Guest to VLAN 50. The guest network is logically separated from the internal departmental VLANs.

Verification

The implementation was verified using Cisco IOS show commands and end-device ping tests. Evidence captured for the portfolio includes VLAN verification, trunk verification, IPv4 verification, IPv6 verification, inter-VLAN testing, and guest isolation tests.

Security and Management

The network uses VLAN segmentation to separate management, production, finance, IT services and guest traffic. Router management access was configured with an enable secret plus console and VTY passwords. The guest network is kept in its own VLAN to provide logical isolation.

Implementation Notes

During implementation, the wireless laptop initially lacked a compatible wireless interface. Installing the WPC300N module allowed the laptop to associate with the guest access point. IPv6 routing was subsequently enabled with the ipv6 unicast-routing command and all router subinterfaces were verified as up/up.

Conclusion

The final design provides a segmented dual-stack network with departmental VLANs, centralised inter-VLAN routing, a dedicated guest WLAN, and documented verification evidence. The Packet Tracer project and configuration evidence form the implementation record for the portfolio.